Deep Hedging under Transaction Costs

CVaR-optimal neural hedging benchmarked against
Black–Scholes delta and no-trade-band strategies

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October 2026

Abstract

We study the hedging of a short European call option under proportional transaction costs. A recurrent neural-network policy, which observes the moneyness, the remaining time to maturity and its current position, is trained end-to-end through a Monte Carlo market simulator to minimise the Conditional Value-at-Risk (CVaR) of the terminal profit and loss. Without costs, the network recovers the Black–Scholes delta from simulated data alone. With 1% proportional costs, it reduces $\mathrm{CVaR}_{95\%}$ out-of-sample by 41% relative to Black–Scholes delta hedging and by 8% relative to the best tuned classical no-trade-band strategy, a gap of roughly twelve times the seed-to-seed standard deviation. We document every building block against theory and discuss the limitations of the simulated setting.

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1 Introduction

A dealer who sells an option is exposed to the movements of the underlying and reduces this exposure by trading the underlying asset. In the frictionless Black–Scholes world the optimal strategy is known in closed form: hold $\Delta = \partial C/\partial S$ shares and rebalance continuously. In practice two assumptions fail. Rebalancing is discrete, which leaves a residual hedging error, and every trade costs money, which makes frequent rebalancing expensive. Under proportional transaction costs there is no exact closed-form optimal strategy; only asymptotic results for small costs are available [6, 4].

Deep hedging [2] replaces the analytical hedging rule by a neural network and trains it directly on simulated market paths to minimise a chosen risk measure of the hedged position. This report reproduces the core findings of that approach and extends the comparison with classical benchmarks. Our contributions are:

1. a fully validated pipeline: market simulator, pricing, hedging engine and training loop, each checked against analytical results;
2. a CVaR-trained recurrent hedging network that accounts for its current position;
3. a comparison not only with naive Black–Scholes delta hedging, but also with two classical cost-aware strategies (constant and Whalley–Wilmott no-trade bands) whose parameters are tuned on the training data;
4. a robustness analysis over several training seeds.

2 Market model

2.1 Geometric Brownian motion

The underlying price follows a geometric Brownian motion

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \tag{1}$$

where μ is the drift, σ the volatility and W_t a standard Brownian motion. Its solution is

$$S_t = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t \Big), \tag{2}$$

so $\log S_t$ is normally distributed with mean $\log S_0 + (\mu - \sigma^2/2)t$ and variance $\sigma^2 t$. The correction $-\sigma^2/2$ ensures $\mathbb{E}[S_t] = S_0 e^{\mu t}$, since for $X \sim \mathcal{N}(m, v)$ one has $\mathbb{E}[e^X] = e^{m+v/2}$.

2.2 Exact discretisation

On a grid $t_k = k\,\Delta t$, $\Delta t = T/N$, paths are generated exactly, without discretisation error:

$$S_{k+1} = S_k \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\, Z_{k+1} \Big), \qquad Z_{k+1} \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{3}$$

The noise scales with $\sqrt{\Delta t}$ because variances of independent increments add up: N steps of variance s^2 must give the total variance $\sigma^2 T$, hence $s^2 = \sigma^2 \Delta t$. Implementation-wise, the log-increments are summed with a cumulative sum along the time axis, which vectorises the simulation over all paths.

2.3 Parameters and validation

Unless stated otherwise we use $S_0 = K = 100$, $\sigma = 0.2$, $\mu = r = 0$, $T = 30/365$ and $N = 30$ daily rebalancing dates. Training uses 100,000 paths (seed 42); all reported results are computed on 100,000 independent test paths (seed 1).

Quantity	Theory	Simulation	Note
$\mathbb{E}[S_T]$, $\mu = 0$	100.000	100.020	difference ≈ 1 standard error
$\mathbb{E}[S_T]$, $\mu = 0.5$	104.195	104.216	checks the drift term
$\mathrm{Var}(\log S_T)$	0.003288	0.003258	difference ≈ 2 standard errors

Table 1: Validation of the simulator against closed-form moments.

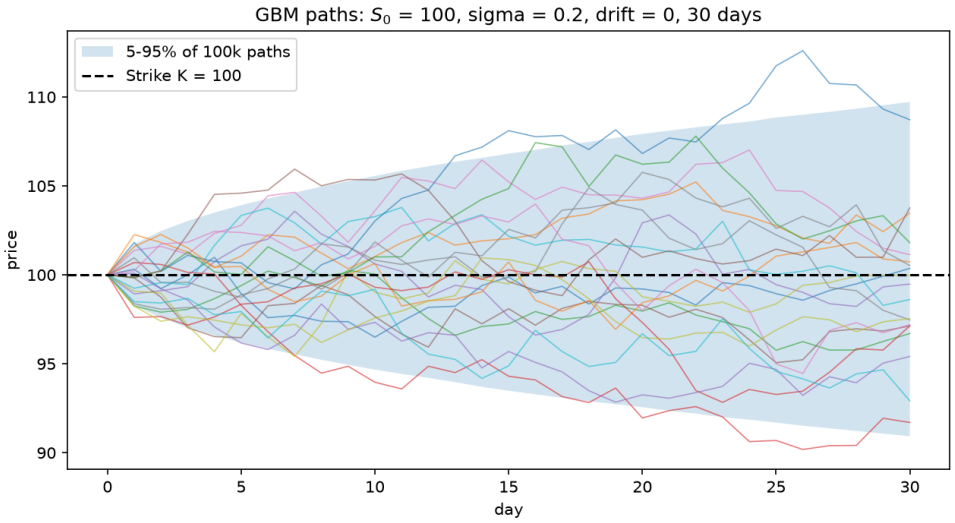


Figure 1: Twenty simulated paths and the 5–95% band of 100,000 paths. The band widens as $\sqrt{t}$ and is slightly skewed upwards, reflecting the log-normal distribution.

3 Pricing and the Black–Scholes delta

With $r = 0$ the fair price of a European call is its expected payoff under the risk-neutral measure, which coincides with the model with $\mu = 0$:

$$C = \mathbb{E}\big[(S_T - K)^+\big] = S\,N(d_1) - K\,N(d_2), \tag{4}$$

$$d_1 = \frac{\ln(S/K) + \tfrac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}, \tag{5}$$

where $\tau = T - t$ is the remaining time to maturity and $N(\cdot)$ is the standard normal distribution function. The Black–Scholes delta and gamma are

$$\Delta = \frac{\partial C}{\partial S} = N(d_1), \qquad \Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{N'(d_1)}{S\,\sigma\sqrt{\tau}}. \tag{6}$$

Setting $r = 0$ is equivalent to working in discounted units and follows [2].

For the at-the-money option of our setup $C = 2.2872$ and $\Delta = 0.5114$. The price matches the Monte Carlo estimate 2.2896 (difference below one quarter of a standard error), and the delta matches the central finite difference $\big(C(S+h) - C(S-h)\big)/(2h)$ to eleven digits.

4 The hedging problem

4.1 Profit and loss

The dealer sells the call for the premium p_0 and holds δ_k units of the underlying on $[t_k, t_{k+1})$, $k = 0, \ldots, N-1$. With proportional costs c per unit of traded notional, the terminal profit and

loss is

$$\mathrm{PnL} = p_0 - (S_N - K)^+ + \sum_{k=0}^{N-1} \delta_k \, (S_{k+1} - S_k) - \sum_{k=0}^{N-1} c \, S_k \, |\delta_k - \delta_{k-1}|, \qquad \delta_{-1} = 0. \tag{7}$$

The four terms are the premium, the option payoff, the gains of the hedge and the trading costs. The turnover $\sum_k |\delta_k - \delta_{k-1}|$ measures the total number of shares traded. We do not charge costs for unwinding the hedge at maturity, since the hedge ratio is then close to 0 or 1.

4.2 Risk measures

Let $L = -\mathrm{PnL}$ denote the loss. Besides the mean and the standard deviation we use

$$\mathrm{VaR}_\alpha(L) = \inf\{\, \ell : \mathbb{P}(L \le \ell) \ge \alpha \,\}, \qquad \mathrm{CVaR}_\alpha(L) = \mathbb{E}\big[\, L \mid L \ge \mathrm{VaR}_\alpha(L) \,\big], \tag{8}$$

the average loss in the worst $(1-\alpha)$ fraction of scenarios, with $\alpha = 0.95$. Unlike the variance, CVaR only penalises the tail of losses and does not penalise gains.

4.3 The Rockafellar–Uryasev representation

Optimising CVaR directly requires sorting, which is inconvenient for gradient-based training. Rockafellar and Uryasev [5] showed that

$$\mathrm{CVaR}_\alpha(L) = \min_{w \in \mathbb{R}} \Big\{ \, w + \frac{1}{1-\alpha} \, \mathbb{E}\big[(L - w)^+\big] \Big\}, \tag{9}$$

with the minimum attained at $w = \mathrm{VaR}_\alpha(L)$. The objective inside the braces is an ordinary expectation, which can be estimated on a mini-batch and differentiated. We treat w as an additional trainable parameter, so that minimising (9) jointly over the network weights and w minimises the CVaR of the hedged position.

4.4 Indifference premium

CVaR is translation equivariant: increasing the premium by Δp reduces CVaR by exactly Δp. The smallest premium at which the hedged position has non-positive CVaR is therefore

$$p^* = p_0 + \mathrm{CVaR}_\alpha(L). \tag{10}$$

A better hedge thus translates directly into a lower price that can be quoted for the same tail risk.

5 Classical benchmarks

5.1 Black–Scholes delta hedging

The naive benchmark holds $\delta_k = \Delta(S_k, \tau_k)$ with $\tau_k = T - t_k$, regardless of costs. Without costs, its discrete hedging error decays as $1/\sqrt{N}$: the error per step has variance of order Δt^2 and the errors are approximately independent, so the total variance is of order $N \Delta t^2 = T^2/N$. With costs, the expected change of the delta per step is of order $\sqrt{\Delta t}$, so the turnover, and with it the trading cost, grows as $N\sqrt{\Delta t} = \sqrt{NT} \propto \sqrt{N}$. Rebalancing more often therefore stops paying off.

Daily delta hedging already removes most of the risk of the short option: the P&L standard deviation falls from 3.46 (unhedged) to 0.36.

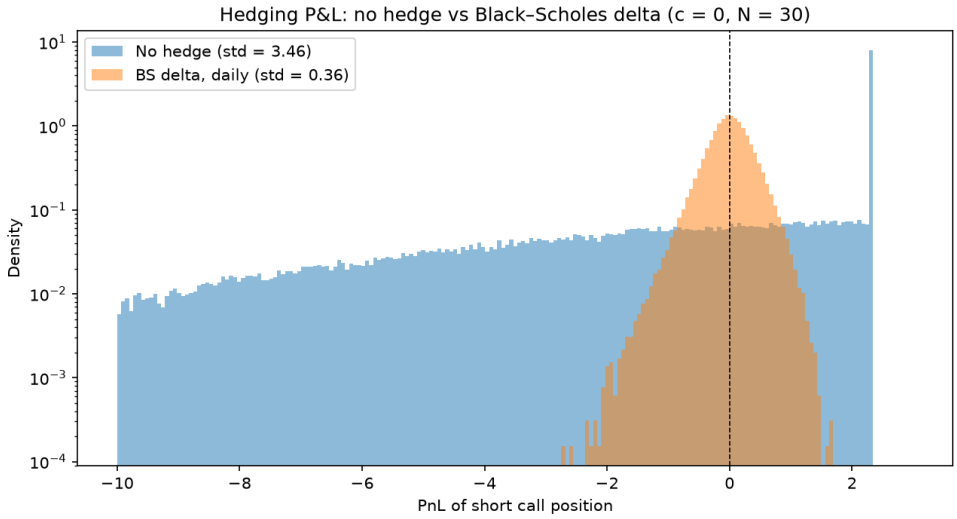


Figure 2: P&L of the short call without hedging and with daily BS delta hedging ($c = 0$, $N = 30$, log scale). Unhedged, the gain is capped at the premium while losses are unbounded.

	$c = 0$		$c = 0.1\%$		$c = 1\%$	
N	mean	std	mean	std	mean	std
5	0.000	0.839	−0.112	0.849	−1.114	0.953
10	0.000	0.608	−0.147	0.619	−1.451	0.767
30	0.000	0.358	−0.222	0.373	−2.224	0.756
100	0.000	0.199	−0.367	0.239	−3.679	1.206

Table 2: Black–Scholes delta hedging versus the number of rebalancing dates N. Without costs, std $\cdot \sqrt{N} \approx 1.9$–$2.0$ is nearly constant. With 1% costs, the mean loss at $N = 100$ exceeds the option premium (2.29) and the risk increases again.

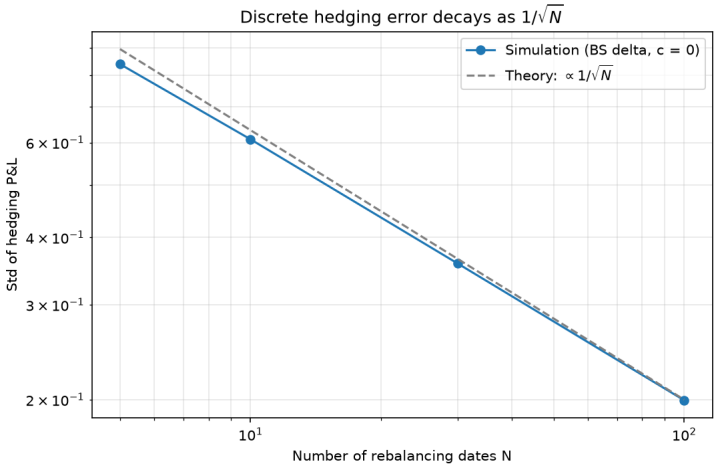


Figure 3: Discrete hedging error of BS delta hedging without costs follows the $1/\sqrt{N}$ law.

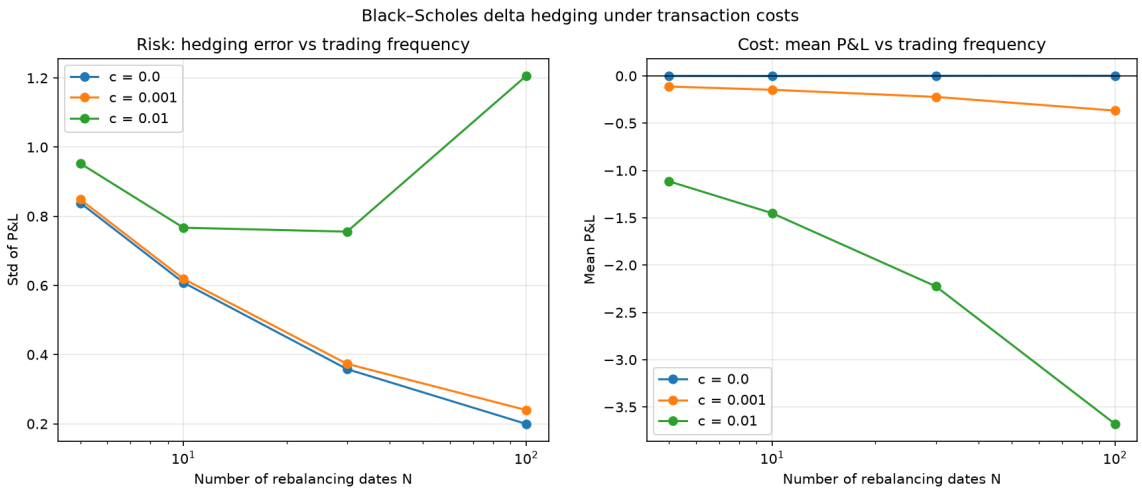


Figure 4: Risk (left) and mean P&L (right) of BS delta hedging versus rebalancing frequency for three cost levels.

5.2 Constant no-trade band

The classical response to proportional costs is a no-trade region around the delta. With half-width h, the position is only adjusted when it leaves the band, and then only to the nearest edge:

$$\delta_k = \min\Big(\max\big(\delta_{k-1},\, \Delta_k - h\big),\, \Delta_k + h \Big). \tag{11}$$

For $h = 0$ this is BS delta hedging. The width h is tuned on the training paths to minimise $\mathrm{CVaR}_{95\%}$; the optimum is $h = 0.11$.

5.3 Whalley–Wilmott band

Asymptotic analysis for small costs [6] suggests a state-dependent half-width that scales with the option gamma:

$$h_k = \kappa \left(c\, S_k\, \Gamma_k^2 \right)^{1/3}, \tag{12}$$

where all model constants (in particular risk aversion) are absorbed into the scale κ, which is again tuned on the training paths ($\kappa = 0.7$).

6 Deep hedging

6.1 Policy

The hedging decision is a function of the observable state,

$$\delta_k = f_\theta\big(x_{1,k},\, x_{2,k},\, \delta_{k-1}\big), \qquad x_{1,k} = \frac{\log(S_k/K)}{\sigma\sqrt{T}}, \qquad x_{2,k} = \frac{\tau_k}{T}, \tag{13}$$

where f_θ is a multilayer perceptron $3 \to 32 \to 32 \to 1$ with ReLU activations (1,217 parameters). Inputs are normalised to order one; the first feature measures the moneyness in units of the total standard deviation, as in d_1. The current position δ_{k-1} is essential under costs: without it the policy cannot tell whether a trade is worth its cost.

Because the position at day k is an input at day $k+1$, the policy is rolled out sequentially over the N dates, which makes the network recurrent. In the cost-free experiments the position input is omitted ($2 \to 32 \to 32 \to 1$, 1,185 parameters) and all dates are evaluated in a single forward pass.

6.2 Training

The policy is trained end-to-end through the simulator. At each iteration a mini-batch of 10,000 paths is drawn, the policy is rolled out, the P&L (7) is computed with differentiable tensor operations, and the loss (9) (or the mean squared P&L in the first experiment) is backpropagated through all N decisions to the network weights and to w. We use Adam [3] with learning rate 10^{-3} for the network and 10^{-2} for w.

6.3 Why not a tree-based model

The problem has no labels: the optimal position at a given date is unknown, and the objective is defined on the whole trajectory, where each decision feeds into the next state. Training therefore requires a policy that is differentiable in its parameters, so that the gradient of the risk measure can be propagated through the simulator. Tree ensembles are piecewise constant and not differentiable in their parameters, so they do not fit this end-to-end formulation. The approach is closest to direct policy optimisation in reinforcement learning, with the difference that here the environment itself is differentiable and gradients are exact.

7 Experiments

7.1 Without costs: recovering Black–Scholes

Trained on the mean squared P&L, the network learns positions that differ from the Black–Scholes delta by 0.0086 on average, out-of-sample. Its P&L standard deviation is 0.369 versus 0.358 for BS delta, and the results on training and test paths coincide, so the network has learned the hedging rule rather than memorising paths.

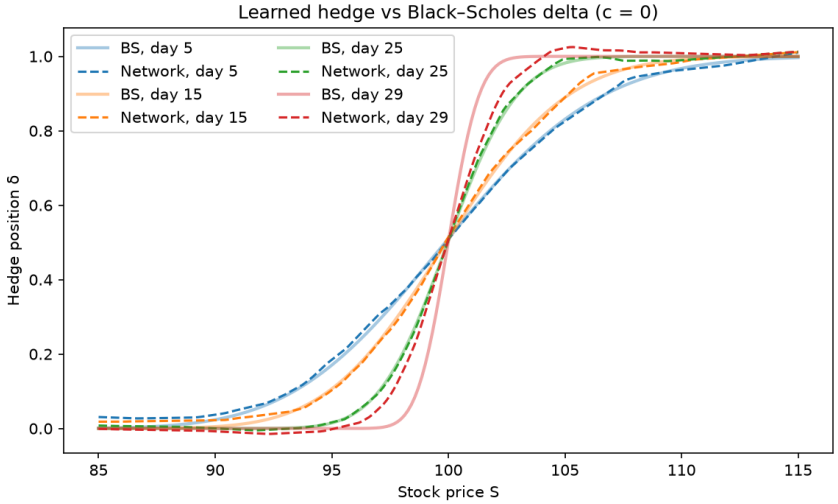


Figure 5: Learned hedge ratio versus the Black–Scholes delta for several dates, without transaction costs.

7.2 Without costs: CVaR versus variance

Strategy	std	$\mathrm{CVaR}_{95\%}$
BS delta	**0.358**	0.833
Network, MSE loss	0.369	0.844
Network, CVaR loss	0.409	**0.739**

Table 3: Out-of-sample comparison without transaction costs.

Each strategy is best on the criterion it targets. The CVaR-trained network shortens the left tail by 11% at the cost of a wider distribution: the Black–Scholes delta is close to variance-optimal but not tail-optimal.

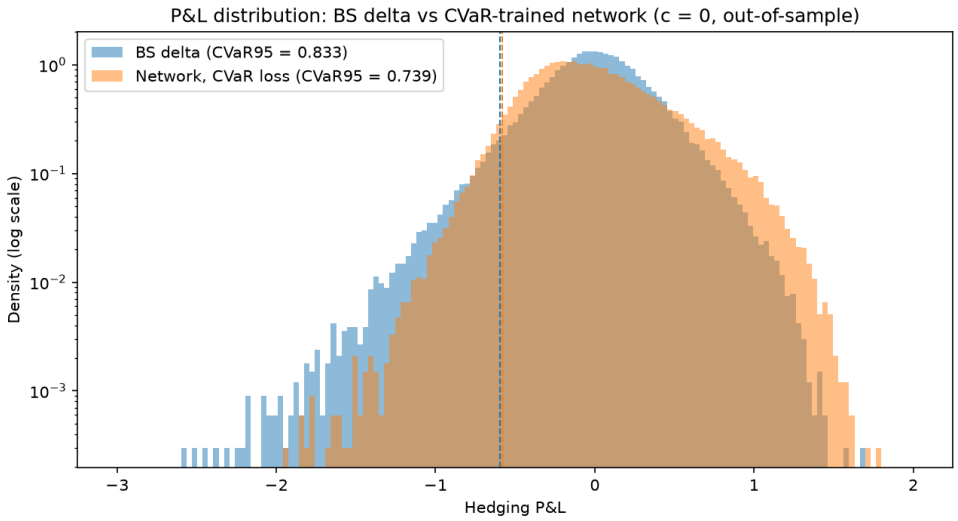


Figure 6: P&L distributions (log scale) of BS delta and the CVaR-trained network; dashed lines mark $-\mathrm{VaR}_{95\%}$.

7.3 With 1% transaction costs

Strategy	mean P&L	std	$\mathrm{CVaR}_{95\%}$	turnover
No hedge	-0.02	3.47	10.06	0.00
BS delta	-2.22	0.75	4.06	2.22
Constant band, $h = 0.11$	-1.04	0.71	2.61	1.03
Whalley–Wilmott band, $\kappa = 0.7$	-1.04	0.78	2.76	1.03
Network (5 seeds)	-1.22 ± 0.02	$\mathbf{0.65 \pm 0.01}$	$\mathbf{2.40 \pm 0.02}$	≈ 1.3

Table 4: Out-of-sample results with $c = 1\%$. Band parameters are tuned on the training paths; network results are mean $\pm$ standard deviation over five training seeds.

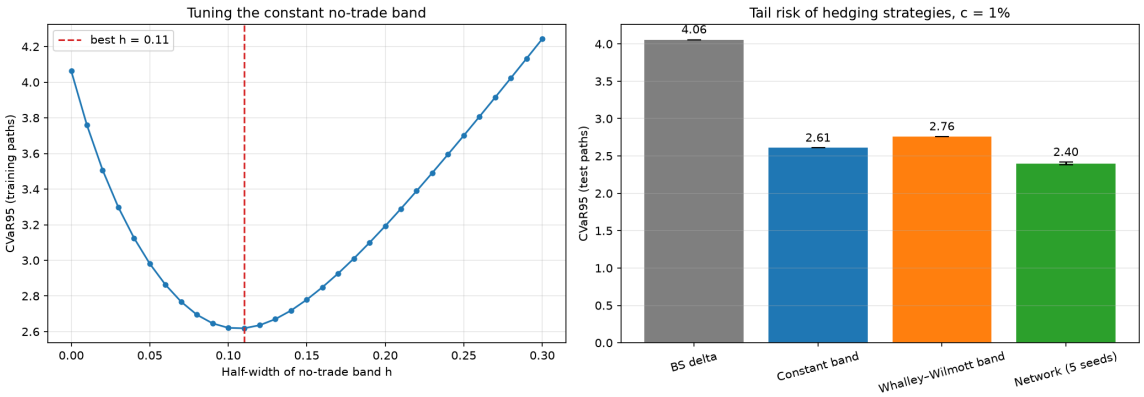


Figure 7: Left: tuning of the constant band width on the training paths. Right: out-of-sample $\text{CVaR}_{95\%}$ of all strategies; the error bar shows the seed-to-seed standard deviation of the network.

The main observations are:

- Most of the improvement over BS delta comes from the no-trade principle itself: a constant band with a single tuned parameter reduces $\text{CVaR}_{95\%}$ by 36%.
- The network improves on the best classical benchmark by a further 8% in $\text{CVaR}_{95\%}$ and 9% in standard deviation. The gap of 0.21 is about twelve times the seed-to-seed standard deviation, and every individual seed beats the band.
- The network accepts a slightly larger mean loss than the band: it trades somewhat more to cut the tail, which is consistent with its objective.
- The Whalley–Wilmott band underperforms the constant band. It is derived for small costs and utility maximisation; under a CVaR objective with 1% costs its widening near expiry, where the gamma of an at-the-money option explodes, leaves the position under-hedged exactly when the tail outcome is decided.
- The indifference premium for the same $\text{CVaR}_{95\%}$ is $2.29 + 2.40 \approx 4.70$ for the network versus $2.29 + 4.06 \approx 6.35$ for BS delta, so the option could be quoted about 26% cheaper.

7.4 What the network learned

Fixing the date and the price and varying only the current position reveals the learned trading rule. Unlike the BS delta, which jumps to the same target from any starting position, the learned rule has a slope well below one in the current position: the network moves only part of the way towards its target and spreads the adjustment over several days, which saves transaction costs. The fixed point of the rule, where no trade occurs, lies close to the BS delta for each price level. This partial rebalancing is a smooth approximation of the band structure of the theoretical solution rather than an exact flat no-trade region. Regions of the state space that are rarely visited during training, such as a zero position at $S = 105$ in mid-life, should be interpreted with caution.

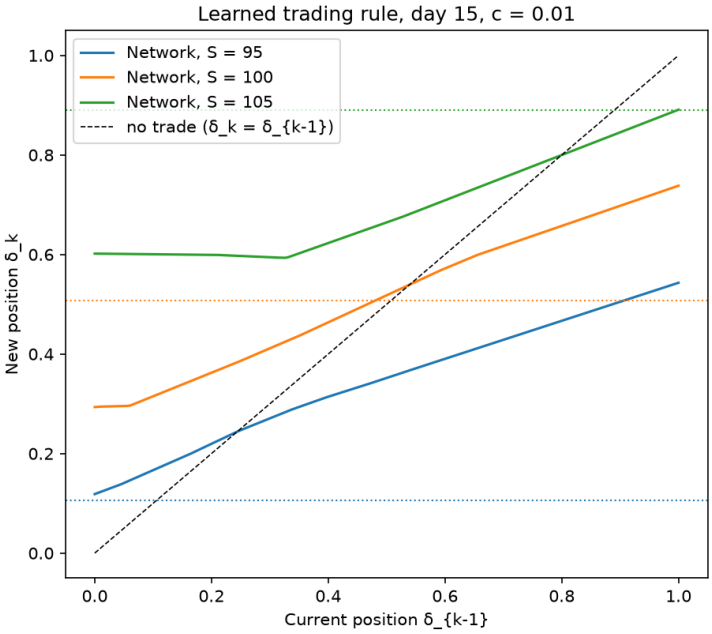


Figure 8: Learned trading rule at day 15 with $c = 1\%$: new position versus current position for three price levels. The diagonal corresponds to no trade; dotted lines show the BS delta.

8 Limitations and model risk

- **Market dynamics.** Training and testing use geometric Brownian motion, which has neither fat tails nor volatility clustering. Results on real markets may differ.
- **Premium.** The option is sold at the frictionless Black–Scholes price rather than at a market quote. Relative comparisons between strategies are unaffected, since the premium is identical for all of them, but absolute profitability is not assessed.
- **Frictions.** Costs are proportional and constant; there is no market impact and no cost for unwinding at maturity.
- **Scope.** A single at-the-money option with one maturity, hedged in the underlying only.

9 Next steps

- Replace GBM by a block bootstrap of historical S&P 500 returns (2005–2017), which preserves fat tails and volatility clustering, and test on realised paths from 2018–2026.
- Add an ex-ante volatility feature to the policy (trailing realised volatility, a GARCH(1,1) forecast with parameters estimated on the training period only, or the VIX), and report results by volatility regime and in historical stress windows.
- Forecast VaR and ES of the hedged book with a neural quantile model and backtest it (exceedance rate, Kupiec test) against historical simulation, linear quantile regression and gradient boosting.

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